

DeepSeek is a generative artificial intelligence chatbot developed by the Chinese company DeepSeek. Released on 20 January 2025, DeepSeek-R1 surpassed ChatGPT as the most downloaded freeware app on the iOS App Store in the United States by 27 January. DeepSeek's success against larger and more established rivals has been described as "upending AI" and initiating "a global AI space race". DeepSeek's compliance with Chinese government censorship policies and its data collection practices have also raised concerns over privacy and information control in the model, prompting regulatory scrutiny in multiple countries. However, it has also been praised for its open weights and infrastructure code, energy efficiency and contributions to open-source artificial intelligence.

History

On 10 January 2025, DeepSeek released the chatbot, based on the DeepSeek-R1 model, for iOS and Android. By 27 January, DeepSeek-R1 surpassed ChatGPT as the most-downloaded freeware app on the iOS App Store in the United States, which resulted in an 18% drop in Nvidia's share price. And after a "large-scale" cyberattack on the same day disrupted the proper functioning of its servers, DeepSeek had limited its new user registration to phone numbers from mainland China, email addresses, or Google account logins.

On 3 April 2025, in collaboration with researchers at Tsinghua University, DeepSeek published a paper unveiling a new model that combines the techniques generative reward modeling (GRM) and self-principled critique tuning (SPCT). The resulting model is referred to as DeepSeek-GRM. The goal of using these techniques is to foster more effective inference-time scaling within their LLM and chatbot services. Notably, DeepSeek has said that these new models will be released and made open source.

On 30 April 2025, Deepseek released its math-focused Artificial Intelligence Model named "DeepSeek-Prover-V2-671B". This model is useful for formal theorem proving and mathematical reasoning.

On 24 April 2026, DeepSeek released DeepSeek V4 and V4-Pro.

Usage

DeepSeek can answer questions, solve logic problems, and write computer programs on par with other chatbots, according to benchmark tests used by American AI companies.

Users can access the chatbot for free through the official DeepSeek website or mobile application, without limitation on the number of queries. DeepSeek only supports user-signup via a global email service, e.g. Gmail, Google or Yahoo. DeepSeek also offers access to the R1 and V3 models that power the chatbot via an API with a usage-based pricing model. This modality is primarily targeted towards developers and businesses. As of February 2025, API usage is priced at approximately \$0.28 per million input tokens and \$0.42 per million output tokens, making it less expensive than some competing services. Its web version is completely free, with 500 messages per hour cap limit to prevent bots from spamming.

Operation

DeepSeek-V3 uses significantly fewer resources compared to its peers. For example, whereas the world's leading AI companies train their chatbots with supercomputers using as many as 16,000 graphics processing units (GPUs), DeepSeek claims to have needed only about 2,000 GPUs—namely, the H800 series chips from Nvidia. It was trained in around 55 days at a cost of US\$5.58 million, which is roughly one-tenth of what tech giant Meta spent building its latest AI technology.

Reactions

DeepSeek's success against larger and more established rivals has been described as "upending AI", constituting "the first shot at what is emerging as a global AI space race", and ushering in "a new era of AI brinkmanship".

Challenge to US AI dominance

DeepSeek's competitive performance at relatively minimal cost has been recognized as potentially challenging the global dominance of American AI models. Various publications and news media, such as The Hill and The Guardian, have described the release of the R1 chatbot as a "Sputnik moment" for American AI, echoing Marc Andreessen's view. OpenAI wrote a letter to the Office of Science and Technology Policy (OSTP), in March 2025, citing issues concerning a possibility that Deepseek could manipulate responses to cause harm.

Chinese perspective

DeepSeek's founder Liang Wenfeng has been compared to OpenAI CEO Sam Altman, with CNN calling him the Sam Altman of China and an evangelist for AI. Chinese state media widely praised DeepSeek as a national asset. On 20 January 2025, Chinese Premier Li Qiang invited Wenfeng to his symposium with experts and asked him to provide opinions and suggestions on a draft for comments of the annual 2024 government work report. On 20 February 2025, Wenfeng met with General Secretary of the Chinese Communist Party Xi Jinping, who encouraged party and state leaders to experiment with DeepSeek. Government officials responded to Xi's approval of the chatbot by reportedly using it to draft legal judgements, propose medical treatment plans, and analyze surveillance videos to search for missing persons.

Performance and success

Leading figures in the American AI sector had mixed reactions to DeepSeek's performance and success. Microsoft CEO Satya Nadella and OpenAI CEO Altman—whose companies are involved in the United States government-backed "Stargate Project" to develop American AI infrastructure—both called DeepSeek "super impressive". Various companies including

Amazon Web Services, Toyota, and Stripe are seeking to use the model in their program. When American President Donald Trump announced The Stargate Project, he referred to DeepSeek as a wake-up call and a positive development.

Other leaders in the AI field, however—including Scale AI CEO Alexandr Wang, Anthropic cofounder and CEO Dario Amodei, and Elon Musk—have expressed skepticism of the app's performance or of the sustainability of its success. Wang in particular referred to DeepSeek-V3 as "earth-shattering" and DeepSeek-R1 as "top performing, or roughly on par with the best American models", but speculated that China may possess more AI-powering Nvidia H100 GPUs than thought.

Stock market implications

DeepSeek's optimization of limited resources has highlighted potential limits of United States sanctions on China's AI development, including export restrictions on advanced AI chips to China. The success of the company's AI models consequently "sparked market turmoil" and caused shares in major global technology companies to plunge on 27 January 2025: Nvidia's stock fell by as much as 17–18%, as did the stock of rival Broadcom. Other tech firms also sank, including Microsoft (down 2.5%), Google's owner Alphabet (down over 4%), and Dutch chip equipment maker ASML (down over 7%). A global sell-off of technology stocks on Nasdaq, prompted by the release of the R1 model, led to record losses of about \$593 billion in the market capitalizations of AI and computer hardware companies; and by the next day a total of \$1 trillion of value was wiped from American stocks.

Concerns

Distillation

DeepSeek has been reported to sometimes claim that it is ChatGPT. OpenAI said that DeepSeek may have "inappropriately" used outputs from its model as training data in a

process called distillation. However, there is currently no method to prove this conclusively.

Censorship

DeepSeek's compliance with Chinese government censorship policies and its data collection practices have raised concerns over information control in the model, prompting regulatory scrutiny in multiple countries.

Reports indicate that it applies content moderation in accordance with the government's "public opinion guidance" regulations, limiting responses on topics such as the Tiananmen Square massacre and Taiwan's political status. DeepSeek models that have been uncensored also display a bias towards Chinese government viewpoints on controversial topics such as Xi Jinping's human rights record and Taiwan's political status. However, users who have downloaded the models and hosted them on their own devices and servers have reported successfully removing this censorship.

Some sources have observed that the official application programming interface (API) version of R1, which runs from servers located in mainland China, uses censorship mechanisms for topics considered politically sensitive for the government of China. For example, the model may initially generate answers to questions about the 1989 Tiananmen Square massacre, persecution of Uyghurs, comparisons between Xi Jinping and Winnie the Pooh, and human rights in China, but a censorship mechanism deletes the uncensored response afterwards and replaces it with a message such as: "Sorry, that's beyond my current scope. Let's talk about something else."

The post hoc censorship mechanisms and restrictions added on top of the model's output can be removed in the open-source version of the R1 model. If the "core Socialist values" defined by the Chinese Internet regulatory authorities are undermined in a response, or if a response undermining the reunification of mainland China and Taiwan is produced, the relevant

response is deleted, but the prompt can be still be edited and resubmitted. Locally hosted instances of R1 are still occasionally reported to provide answers consistent with Chinese Communist Party propaganda narratives. When tested by NBC News, for instance, DeepSeek's R1 initially described Taiwan as "an inalienable part of China's territory", and stated:"We firmly oppose any form of 'Taiwan independence' separatist activities and are committed to achieving the complete reunification of the motherland through peaceful means."

In January 2025, however, Western researchers were able to trick DeepSeek into giving certain answers to some of these topics by requesting in its answer to swap certain letters for similar-looking numbers.

In July 2025, Christo Grozev, using R1 for text processing and by asking it not to change his text, found that the chatbot had modified his text and parroted Russian propaganda.

Security and privacy

Some experts fear that the government of China could use the AI system for foreign influence operations, spreading disinformation, surveillance, and the development of cyberweapons. An article in Wired commented that the DeepSeek online service sending data to its home country might set "the stage for greater scrutiny".

In February 2025, South Korea's data protection regulator, the Personal Information Protection Commission (PIPC), raised concerns over DeepSeek, confirming that it sent the nation's user data to the owner of TikTok (ByteDance) in China. Accordingly, the PIPC banned new downloads of the app until DeepSeek addressed these concerns. The company's representative in South Korea has partially acknowledged its shortcomings in complying with local data protection laws.

On 29 January 2025, cybersecurity researchers from Wiz Research found a significant data breach at DeepSeek. This occurred due to an improper configuration of a cloud storage database that was left available online without any security precautions. The database at

oauth2callback.deepseek.com:9000 and dev.deepseek.com:9000 contained more than a million sensitive entries, which comprised chat messages, API keys, operational details from the backend, system logs, and internal metadata from 6 January 2025. The breach was consequential as it provided complete control over the database, which could lead to unauthorized access and increased privileges within DeepSeek's systems. After Wiz Research reported the issue, DeepSeek secured the database in less than an hour, eliminating any immediate risks. However, the exposure sparked serious concerns about DeepSeek's data security practices and the possibility of facing regulatory action under laws like GDPR or CCPA, particularly if the data included personal information from EU or US residents.

In April 2025, South Korea's Personal Information Protection Commission announced that DeepSeek had transferred personal information on over a million South Koreans to China without consent. In June 2025, the Italian Competition Authority opened an investigation into DeepSeek over false information risks. In September 2025, Crowdstrike found that DeepSeek produces less secure code for groups that the Chinese government disfavors.

Miscellaneous

The Wall Street Journal reported that the DeepSeek app produces instructions for self-harm and dangerous activities more often than its American competitors. Security researchers have found that DeepSeek sends data to a cloud platform affiliated with ByteDance.

In February 2025, sources claimed that DeepSeek began considering raising external funding for the first time, with Alibaba and Chinese state funds expressing interest in investing in DeepSeek. On 21 February, DeepSeek announced plans to release key codes and data to the public starting the following week.

Restrictions

Many countries have raised concerns about data security and DeepSeek's use of personal

data. On 28 January 2025, the Italian data protection authority announced that it is seeking additional information on DeepSeek's collection and use of personal data. On the same day, the United States National Security Council announced that it had started a national security review of DeepSeek, and the United States Navy instructed all its members not to use DeepSeek due to "security and ethical concerns." On 30 January 2025, Italy's data protection authority ordered DeepSeek to block its chatbot in the country after it said the Chinese artificial intelligence startup failed to address the regulator's concerns over its privacy policy. Italian users who downloaded DeepSeek on mobile devices before the ban reported being able to use the chatbot and having access to the web version of DeepSeek after the ban.

Three days later, South Korea's Personal Information Protection Commission opened an inquiry into DeepSeek's use of personal information; the Dutch Data Protection Authority launched an investigation of DeepSeek; Taiwan's digital ministry advised its government departments against using the DeepSeek service to "prevent information security risks"; and Texas Governor Greg Abbott issued a state ban on government-issued devices for DeepSeek, along with Xiaohongshu and Lemon8.

In February 2025, access to DeepSeek was banned on the New South Wales Department of Customer Service's devices. That same month, Australia, South Korea, and Canada banned DeepSeek from government devices, and South Korea suspended new downloads of DeepSeek due to risks of personal information misuse. In March 2025, the U.S. Department of Commerce prohibited DeepSeek on their government devices, due to the alleged "danger of disclosing proprietary information with the Chinese Communist Party". In July 2025, the government of the Czech Republic banned all DeepSeek products in state administration due to "cybersecurity concerns".

See also

HKChat

List of chatbots

Zhejiang University

Notes

References

External links

Official website